

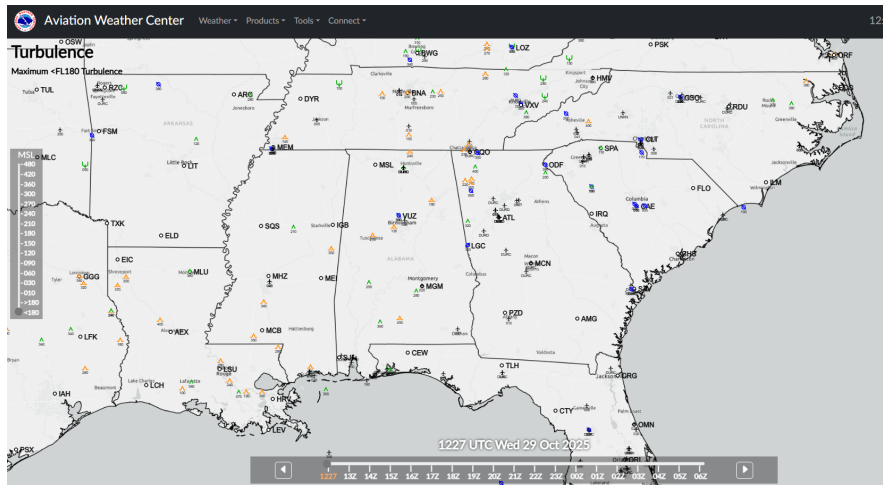
UNIT 4: Weather Analysis Products

c. Pilot Reports & Terminal Aerodrome Forecasts

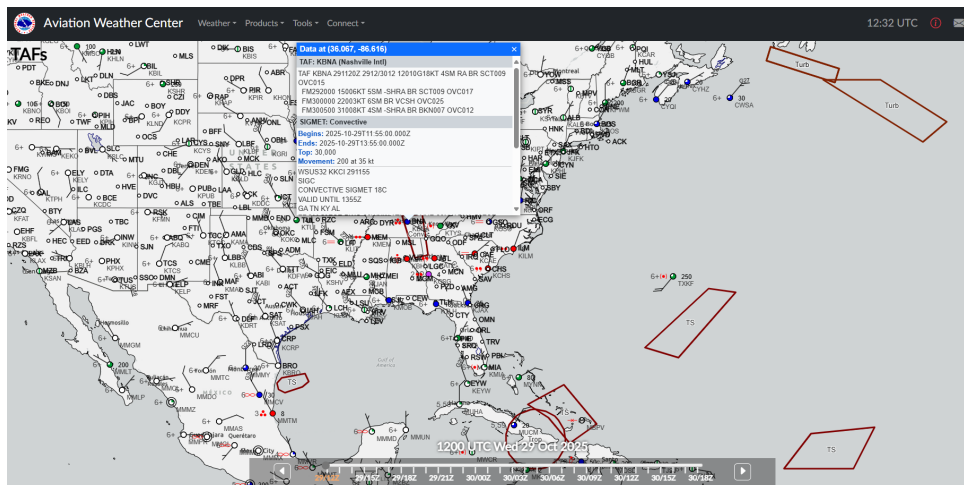
the title slide of 4c needs to have TAFs added

Introduction to Pilot Reports (PIREPS) & Terminal Aerodrome Forecasts (TAFs)

PIREPS are used to record and report in-flight weather occurrences. PIREP data may be obtained from a pilot either in the air or on the ground, or from a reliable source on the ground.



TAFs are used to provide a forecast for the most probable weather elements expected during the forecast period for a specific location. Operationally, time periods vary between the services. Air Force units issue a 30-hour forecasts while Navy and Marine Corps units issue a 24-hour forecasts. The forecast location is a five-statute mile radius extending outward from each location.



The goal of this objective will be to show you have the capability to decode PIREPS and TAFS. To successfully demonstrate competency on this topic, you will be given information through lecture, a series of questions and tasks designed to display your ability to correctly interpret encoded weather data discussed throughout this objective. These foundational analysis concepts will be measured with a Progress Check (PC) at the end of the objective and must be passed to continue onto the next objective. Follow your instructor's guidance as they provide information and instruction for completing all requirements necessary to pass this objective.

INPUT PIREP/TAF SOPS HERE

Reference Manual

Like the publication containing all information and guidance concerning surface weather observations, a manual also exists outlining how to properly Decode PIREPS and TAFs. For this course, all branches will use the air force publication **AFMAN 15-124** for guidance on PIREPS and TAFs.

Exercise / Practice: Navigating the Manual

Decoding PIREPS

Each PIREP must contain:

- Message Type
- Location
- Time
- Flight Level
- Type of Aircraft
- At least one other element to include either sky cover, present weather, temperature, flight level, wind direction and speed, turbulence, or icing.



Station Identifier

The airfield identifier of the unit transmitting the pilot's observation. This identifier will be the same as the identifier used in that station's observations.

Message Type

There are two message types: UA (routine report) or UUA (urgent) report.

Location

The point (/OV) where a pilot observed weather phenomena. It has different formats. For single location reports, the location can be an airport identifier to include during climb (DURC) away from the airfield or during descent (DURD) to the airfield. It can also be a latitude/longitude coordinate such as 2139N 06525W. Pilots can also report weather between two locations using airfield identifiers, latitude/longitude coordinates, or a combination of both.

Time

The UTC time (/TM) pilot observed the weather phenomena.

Flight level

The aircraft's altitude (/FL) in hundreds of feet is like METAR cloud heights. Decode /FLUNKN as flight level unknown.

Type of aircraft

Read the aircraft type (/TP) without any modification.

Sky cover

The sky cover (/SC) at the pilot's flight level. Each layer has an amount as well as its base and top in hundreds of feet, if known. Overcast layers may have an unknown base or top.

Weather

The weather (/WX) group contains the flight visibility (FV) indicator, visibility in two digits, its units of measure, and any weather phenomenon. Decode visibility in whole miles with two exceptions, unrestricted visibility (99) and visibility less than 1/2 mile (00).

The units of measure may be statute miles or kilometers. If statute miles, SM will follow the visibility values. There is no indicator for kilometers which are normally only at non-United States locations overseas.

Temperature

Decode temperature (/TA) in whole degrees Celsius. If the letter M precedes the temperature, decode it as a negative number.

Winds

The wind (/WV) group has six digits. The first three are direction and the last three are the speed.

Turbulence

The turbulence (/TB) group has four parts: duration, intensity, type, and altitude. The three durations, four intensities, and two types combine to describe a multitude of turbulence scenarios. For example, INTMT LGT CHOP is intermittent light chop turbulence and MOD OCNL SEV CAT indicates moderate occasional severe clear air turbulence.

Duration:

- OCNL – Occasional
- INTMT – Intermittent
- CONS – Continuous

Intensity:

- LGT – Light
- MOD – Moderate
- SEV – Severe
- EXTRM – Extreme

Type:

- CAT – Clear Air Turbulence
- CHOP - Chop

Altitude has three formats, and you decode each one differently. First, when the PIREP has turbulence without an altitude, the turbulence is at the aircraft's flight level. Second, it may have the base and top of turbulence depicted in hundreds of feet using three digits. Third, the contractions ABV (above) or BLO (below) may be included with one altitude. For example:

- /TB INTMT MOD = Intermittent moderate turbulence at flight level.
- /TB LGT CHOP 030-050 = light chop turbulence between 3,000 and 5,000 feet.
- /TB LGT-MOD BLO 080 = Light to moderate turbulence below 8,000 feet.

The last turbulence entry is NEG, meaning the aircraft is in a forecast area of turbulence, but it did not experience any turbulence.

Icing

The icing (/IC) group has four intensities and three types. Decode icing the same as the turbulence group to include “NEG” icing.

Intensity:

- TRACE – Trace (amount too minimal to be properly identified)
- LGT - Light
- MOD - Moderate
- SEV – Severe

Type:

- RIME - Rime
- CLR - Clear
- MXD – Mixed

Exercise / Practice: Decoding PIREPS

Decoding TAFs

The TAF will show weather elements onset, duration, and intensity throughout the forecast period. TAF code nearly mirrors the METAR code with some exceptions. All TAFs will have weather elements in the following order from beginning to end:

- Valid time
- Winds
- Visibility
- Present weather
- Sky condition
- Wind Shear
- Icing
- Turbulence
- Pressure

Structure

A TAF may be multiple lines, or a single line forecast that includes all the elements above. A single line TAF typically occurs when the air mass does not change so individual elements do not change or cross specification criteria throughout the duration of the forecast period.

TAF KBIX 291200Z 2912/3018 13010KT 9999 SCT040 BKN080 WS020/22035KT
QNH2998INS
TEMPO 2918/2922 24025G35KT 6000 TSRA SCT015 BKN030CB
BECMG 3001/3002 27015KT 9999 NSW BKN015 OVC025 520306 620406 QNH2982INS
BECMG 3013/3014 32020KT 9999 OVC015 620306 QNH2965INS
TX22/2919Z TN13/3009Z

Issue and Valid Times

The TAFs issue time tells when the TAF was submitted. The issue time tells the day, hour, and minute the TAF was issued. The valid time tells you the starting time the TAF is forecasting for and the end of the forecast.

All times are encoded using Coordinated Universal Time (UTC) in day, hours, and minutes followed by the letter Z. All valid periods consist of a beginning time and ending time in whole hours. For example, 1420/1520Z is a valid period beginning on the 14th at 2000Z through the 15th at 2000Z.

TAF KBIX **291200Z 2912/3018** 13010KT 9999 SCT040 BKN080 WS020/22035KT
QNH2998INS
TEMPO 2918/2922 24025G35KT 6000 TSRA SCT015 BKN030CB
BECMG 3001/3002 27015KT 9999 NSW BKN015 OVC025 520306 620406 QNH2982INS
BECMG 3013/3014 32020KT 9999 OVC015 620306 QNH2965INS
TX22/2919Z TN13/3009Z

Change Groups

The three change groups are becoming (BECMG), temporary (TEMPO), and from (FM). They show changes from the predominant forecast condition at a specified period.

Becoming (BECMG): This is a predominate change in one or more weather elements. The change typically occurs over a one-hour period. It is the primary forecast change group for the Air Force. It includes all weather elements required in the TAF whether they are changing or not. Navy and Marine Corps TAFs only have the changing element or elements. A BECMG group has a two-digit date with beginning and ending times in whole hours. The elements become the predominant condition after the end time.

From (FM): This is also a predominate change group. It is the primary forecast group for Navy and Marine Corps TAFs. FM groups specify the date, hour, and minutes for the change. 021420Z indicates a change on the 2nd at 1420Z.

Temporary (TEMPO): TEMPO groups list elements that fluctuate between specific criteria thresholds. The key to TEMPO groups is the elements listed NEVER become the predominate weather condition. TEMPO group times are bound by the fluctuating weather element(s). The beginning time fluctuations begin to the time the fluctuations end. It has the same format as the TAF valid time.

TAF KBIX 291200Z 2912/3018 13010KT 9999 SCT040 BKN080 WS020/22035KT
QNH2998INS
TEMPO 2918/2922 24025G35KT 6000 TSRA SCT015 BKN030CB
BECMG 3001/3002 27015KT 9999 NSW BKN015 OVC025 520306 620406 QNH2982INS
BECMG 3013/3014 32020KT 9999 OVC015 620306 QNH2965INS
TX22/2919Z TN13/3009Z

Wind Group

The prevailing wind direction is encoded in three digits to nearest 10 degrees and the speed in two or three digits to the nearest whole knot. Wind gusts will be encoded the same as METAR code.

TAF KBIX 291200Z 2912/3018 **13010KT** 9999 SCT040 BKN080 WS020/22035KT
QNH2998INS
TEMPO 2918/2922 **24025G35KT** 6000 TSRA SCT015 BKN030CB
BECMG 3001/3002 **27015KT** 9999 NSW BKN015 OVC025 520306 620406 QNH2982INS
BECMG 3013/3014 **32020KT** 9999 OVC015 620306 QNH2965INS
TX22/2919Z TN13/3009Z

Visibility, Weather, and Obscuration Group

Visibility is encoded in meters using the METAR observation code's reportable values. Anytime the visibility is below 9999 meters the TAF must include a restrictor to visibility. Precipitation and obscurations to vision use the same coding rules as METAR code. Only one group is normal in a TAF, but multiple groups are possible. No significant weather (NSW) is unique to the TAF

code. It signifies the end of a restricting element to include any feature in the vicinity. Either the restrictor stopped, or it moved away from the forecast location.

TAF KBIX 291200Z 2912/3018 13010KT **9999** SCT040 BKN080 WS020/22035KT
 QNH2998INS
 TEMPO 2918/2922 24025G35KT **6000 TSRA** SCT015 BKN030CB
 BECMG 3001/3002 27015KT **9999 NSW** BKN015 OVC025 520306 620406 QNH2982INS
 BECMG 3013/3014 32020KT **9999** OVC015 620306 QNH2965INS
 TX22/2919Z TN13/3009Z

Sky Condition Group

Sky Condition is encoded in the same format as METAR code. All forecast cloud layers will be arranged from the lowest to the highest cloud up to the first overcast layer. The only cloud type appended to a cloud group is cumulonimbus (CB).

TAF KBIX 291200Z 2912/3018 13010KT 9999 **SCT040 BKN080** WS020/22035KT
 QNH2998INS
 TEMPO 2918/2922 24025G35KT 6000 **TSRA SCT015 BKN030CB**
 BECMG 3001/3002 27015KT 9999 NSW **BKN015 OVC025** 520306 620406 QNH2982INS
 BECMG 3013/3014 32020KT 9999 **OVC015** 620306 QNH2965INS
 TX22/2919Z TN13/3009Z

Low-Level Wind Shear

Wind shear is the change in wind speed and/or direction between surface and 2,000 ft AGL. While some degree of wind shear is almost always present in the atmosphere, strong wind shear can pose a significant hazard to aviation and influence severe weather. Wind shear is encoded in hundreds of feet with the wind direction and speed above the shear height. For example, you encode WS015/12038KT for LLWS at 1,500 feet with winds from 120 degrees at 38 knots above 1,500 feet.

TAF KBIX 291200Z 2912/3018 13010KT 9999 SCT040 BKN080 **WS020/22035KT**
 QNH2998INS
 TEMPO 2918/2922 24025G35KT 6000 TSRA SCT015 BKN030CB
 BECMG 3001/3002 27015KT 9999 NSW BKN015 OVC025 520306 620406 QNH2982INS
 BECMG 3013/3014 32020KT 9999 OVC015 620306 QNH2965INS
 TX22/2919Z TN13/3009Z

Turbulence

Air Force TAFs only encode turbulence between the surface and 10,000 feet (AGL) while Navy and Marine Corps TAFs encode turbulence for all levels between the surface and tropopause. The turbulence group has six digits. The first digit is the group indicator (5) and the second digit is a coded figure for turbulence intensity. The third through fifth digits are the base height of the turbulence layer in hundreds of feet and the sixth digit is the thickness of the layer in thousands of feet.

TAF KBIX 291200Z 2912/3018 13010KT 9999 SCT040 BKN080 WS020/22035KT
 QNH2998INS
 TEMPO 2918/2922 24025G35KT 6000 TSRA SCT015 BKN030CB
 BECMG 3001/3002 27015KT 9999 NSW BKN015 OVC025 520306 620406 QNH2982INS
 BECMG 3013/3014 32020KT 9999 OVC015 620306 QNH2965INS
 TX22/2919Z TN13/3009Z

Icing

Icing is only included if it is not associated with thunderstorms. Air Force TAFs only encode icing between the surface and 10,000 feet (AGL) while Navy and Marine Corps TAFs encode icing for all levels between the surface and tropopause. The icing group has six digits. The first digit is the group indicator (6), and the second digit is a coded figure for the icing intensity and type. The third through fifth digits are the base height of the icing layer in hundreds of feet and the sixth digit is the thickness of the layer in thousands of feet.

TAF KBIX 291200Z 2912/3018 13010KT 9999 SCT040 BKN080 WS020/22035KT
 QNH2998INS
 TEMPO 2918/2922 24025G35KT 6000 TSRA SCT015 BKN030CB
 BECMG 3001/3002 27015KT 9999 NSW BKN015 OVC025 520306 620406 QNH2982INS
 BECMG 3013/3014 32020KT 9999 OVC015 620306 QNH2965INS
 TX22/2919Z TN13/3009Z

Altimeter Setting Group

The altimeter setting (ALSTG) is encoded in inches of mercury. For a single line TAF, the ALSTG will be the lowest expected value for the entire forecast period. Multiple line TAFs encode the lowest ALSTG for the valid time of each predominate group. The group format begins with its indicator (QNH) followed by the lowest ALSTG in four digits without a decimal point then concludes with “INS” for inches of Mercury.

TAF KBIX 291200Z 2912/3018 13010KT 9999 SCT040 BKN080 WS020/22035KT
 QNH2998INS
 TEMPO 2918/2922 24025G35KT 6000 TSRA SCT015 BKN030CB
 BECMG 3001/3002 27015KT 9999 NSW BKN015 OVC025 520306 620406 QNH2982INS
 BECMG 3013/3014 32020KT 9999 OVC015 620306 QNH2965INS
 TX22/2919Z TN13/3009Z

Maximum and Minimum Temperature

Maximum and minimum temperatures are encoded in the last line of the TAF. This remark is to display the forecasted date/time of the maximum and minimum temperatures for ONLY the first 24-hours of the TAF. Its format includes two temperature indicators: TX for the maximum temperature and TN for the minimum temperature. Temperatures are two digits in whole degrees Celsius and preceded with the letter “M” for negative temperatures. Following each temperature is the two-digit date and time to nearest whole hour followed with the letter “Z”.

TAF KBIX 291200Z 2912/3018 13010KT 9999 SCT040 BKN080 WS020/22035KT
QNH2998INS
TEMPO 2918/2922 24025G35KT 6000 TSRA SCT015 BKN030CB
BECMG 3001/3002 27015KT 9999 NSW BKN015 OVC025 520306 620406 QNH2982INS
BECMG 3013/3014 32020KT 9999 OVC015 620306 QNH2965INS
TX22/2919Z TN13/3009Z

Exercise / Practice: Decoding TAFs

Practice: PIREPS & TAFS

PC: Decoding PIREPS & TAFs